Exercise 5.4

I. Construct the following quadrilaterals with the given measurements and also find their area.

1. ABCD, AB = 5 cm, BC = 4.5 cm, CD = 3.8 cm, DA = 4.4 cm and AC= 6.2 cm.
2. PLAY, PL= 7 cm, LA = 6 cm, AY= 6 cm, PA = 8 cm and LY = 7 cm.
3. PQRS, PQ=QR= 3.5 cm, RS= 5.2 cm, SP = 5.3 cm and $\angle Q = 120°$.
4. MIND, MI =3.6 cm, ND = 4 cm, MD= 4 cm, $\angle M = 50°$ and $\angle D = 100°$.
5. AGRI, AG= 4.5 cm, GR = 3.8 cm, $\angle A = 60°$, $\angle G = 110°$ and $\angle R = 90°$.

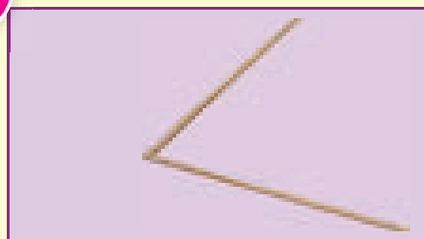
II. Construct the following trapeziums with the given measures and also find their area.

1. AIMS with $\overline{AI} \parallel \overline{SM}$, AI=6*cm*, IM=5*cm*, AM=9*cm* and MS=6.5*cm*.
2. CUTE with $\overline{CU} \parallel \overline{ET}$, CU=7*cm*, $\angle UCE = 80°$ CE=6*cm* and TE=5*cm*.
3. ARMY with $\overline{AR} \parallel \overline{YM}$, AR=7*cm*, RM=6.5*cm* $\angle RAY = 100°$ and $\angle ARM = 60°$
4. CITY with $\overline{CI} \parallel \overline{YT}$, CI=7*cm*, IT=5.5*cm*, TY=4*cm* and YC=6*cm*.

5.12 Construction of Special Quadrilaterals

Before we begin to learn constructing certain quadrilaterals, it is essential to recall their basic properties that would help us during the process. We will try to do this by performing some activities and then sum them up.

Activity



1. Place a pair of *unequal* sticks (say pieces of broomstick) such that they have their end points joined at one end.



2. Now place another such pair meeting the free ends of the first pair.

What is the figure enclosed? It is a quadrilateral. Name it as ABCD. How many sides are there? What are its diagonals? Are the diagonals equal? Are the angles equal?

In the above activity can you get a quadrilateral in which

(i) All the four angles are acute.

(ii) One of the angles is obtuse.

(iii) Two of the angles are obtuse.

(iv) One of the angles is a right angle.

(v) Two of the angles are right angles.

(vi) The diagonals are mutually $\perp^r$.

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